FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2019

PAPER –1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. C

Sol. As V increases, E increases, λ_{min} decreases, $\lambda_{K\alpha}$ remains constant SO $(\lambda_{K\alpha} - \lambda_{min})$ increases

2. A, B, C, D

3. A, B, D

Sol. Work done by B $= \int_0^{u/a} k\left(ut - \frac{1}{2}at^2\right)udt = \frac{ku^4}{3a^2}$

Work done by A $= -\int_0^{u/a} k\left(ut - \frac{1}{2}at^2\right)atdt = -\frac{5ku^4}{24a^2}$

Energy stored in spring $= \frac{1}{2}k\left(\frac{u^2}{2a}\right)^2 = \frac{ku^4}{8a^2}$

4. A, B, C

Sol. (A) $a = \frac{m_1 = m_2}{m_1 + m_2}g \Rightarrow \frac{g}{5} = 2m/s^2$

$h = \frac{1}{2}at^2 \Rightarrow t = 2\,sec.$

(B) $H = 4 + \frac{U^2}{2g}$ where $U = at$

$= 4.8$ m.

(C) 6 kg block will be lifted after time $2 + \frac{2U}{g} = 2.8\,sec.$

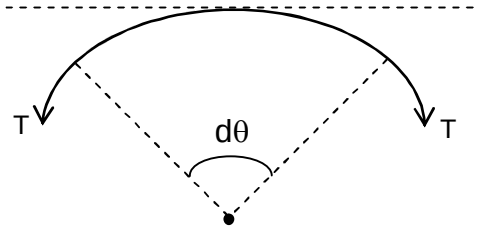
5. A, B, D

6. A, B, C, D

7. B

Sol. $2T\sin\frac{d\theta}{2} = Rd\theta ap\omega^2 R\left[\sin\frac{d\theta}{2} \approx \frac{d\theta}{2}\right]$

$T = apR^2\omega^2$.



8. A

Sol. $KE = \frac{1}{2}m\omega^2R^2 = \pi R^3 ap\omega^2$.

$\text{Stress in ring} = \frac{T}{a} = pR^2\omega^2$

$PE = \frac{1}{2}\frac{(\text{stress})^2}{Y} \times \text{volume}$

$PE = \frac{\pi p^2R^5a\omega^4}{y}$

9. B

Sol. Finally, the capacitors are in parallel and total charge (= q_0) distributes between them in direct ratio of capacity.

$\therefore \quad q_{C_2} = \left(\frac{C_2}{C_1 + C_2}\right)q_0 \rightarrow$ in steady state.

But this charge increases exponentially.

Hence, charge on C_2 at any time t is

$q_{C_2} = \left(\frac{C_2 q_0}{C_1 + C_2}\right)\left(1 - e^{-t/\tau_C}\right)$

Initially,C_2 is uncharged so, whatever is the charge on C_2 it is charge flown through switches

10. A

Sol. Common potential in steady state when they finally come in parallel is

$V = \frac{\text{Total charge}}{\text{Total capacity}} = \frac{q_0}{C_1 + C_2}$

Total heat dissipated $U_i - U_f$

$= \frac{q_0^2}{2C_1} - \frac{1}{2}(C_1 + C_2)\left(\frac{q_0}{C_1 + C_2}\right)^2$

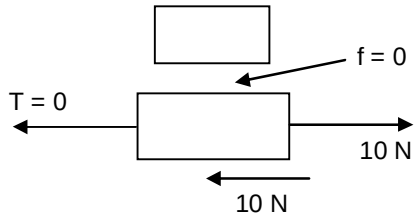
$= \left(\frac{q_0^2}{2C_1}\right)\left(\frac{C_1C_2}{C_1 + C_2}\right)^2$

SECTION – D

11. 00000.00

Sol. Since force which try to move the system is less than max friction hence system does not move.

Hence $T = 0, f = 0$



12. 00002.00

Sol. $$n = \frac{\left(\frac{F}{A}\right)}{\left(\frac{dv}{dx}\right)} = \frac{\frac{0.02}{10}}{\frac{2}{1}} = 10^{-3}\,kg/ms = 10^{-2}\,poise$$

13. 00005.00

Sol. Velocity of bike $V_b = 30\ m/s$

Let velocity of car be V_c

$$f_{obs} = f_{actual} \times \frac{V_{so} + V_b}{V_{so} - V_c}$$

$$120\ Hz = 100\ Hz \times \frac{330 + 30}{330 - V_c}$$

$\Rightarrow \quad V_C = 30\ m/s$

Distance between initial and final pulse of horn

$= V_{so} \times \Delta t - V_C\ \Delta t \qquad \{\Delta t \text{ is duration of emission of sound}\}$

$= (330 - 30) \times 6 = 1800\ m$

This wave train travels with 330 m/s and cross bike rider moving with velocity of 30 m/s towards it.

$$\text{time taken} = \frac{\text{length of wave train}}{V_{so} + V_b} = \frac{1800}{330 + 30} = \frac{180}{36} = 5\,sec$$

14. 00500.00

Sol. EMF, ε_1 (due to varying B) $= B_0 l\ vt$

ε_2 (due to motion of PQ) $= B_0 l\ vt$.

$\varepsilon = \varepsilon_1 + \varepsilon_2 = 2B_0\ l\ vt$

$\because \quad F = ilB = \frac{2B_0 lvt}{R} lB_0 t = \frac{2B_0^2 l^2 vt^2}{R}$

15. 00025.00

Sol. $(V_{rms})_R = i_{rms} R = 150\ V$

$(V_{rms})_L = \sqrt{V_{rms}^2 - (V_{rms})_R^2} = 200\ V$

$(V_{rms})_L = i_{rms} X_L = i_{rms}\, 2\pi f L$

$$L = \frac{200}{10 \times 2\pi \times 50} = \frac{1}{5\pi} H$$

16. 00004.00

Sol. $|\varepsilon| = |\vec{v}.(\vec{B}\times\vec{l})|$

$$= (2\hat{i})\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 1 & -2\\ 0 & 3/5 & 4/5\end{vmatrix} = 4$$

17. 00450.00

Sol. $V_1 = \dfrac{(2C)V}{3C} = \dfrac{2}{3}V$

$U_1 = \dfrac{1}{2}C\dfrac{4}{9}V^2 = \dfrac{2CV^2}{9}$

$V_2 = \dfrac{V}{3}$

$U_2 = \dfrac{1}{2}(2C)\dfrac{V^2}{9} = C\dfrac{V^2}{9} = \dfrac{CV^2}{9}$

Initial total energy $= \dfrac{CV^2}{3}$

Final total energy $= \dfrac{1}{2}(2C)V^2 = CV^2$

Charge flow through the battery $= 2CV - \dfrac{2}{3}CV = \dfrac{4}{3}CV$

$\Rightarrow W_{battery} = \dfrac{4}{3}CV^2$

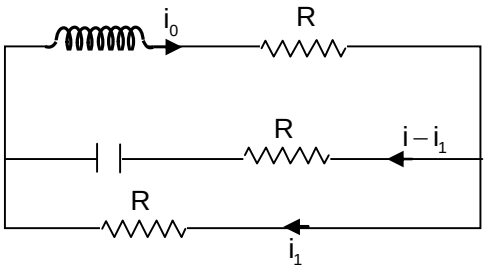
$\Rightarrow$ Heat dissipated $= \dfrac{4CV^2}{3} - \dfrac{2CV^2}{3} = \dfrac{2}{3}CV^2$

18. 00000.00

Sol. $R = \sqrt{\dfrac{L}{C}} \Rightarrow RC = L/R$ i.e. time constants of RC & LR circuit are same.

After switching from 1 to 2

$i_0 = \dfrac{\xi}{2R}$



From loop rule,

$$-(i_0 - i_1)R + \varepsilon/2 + i_1 R = 0$$

$\Rightarrow i_1 = \dfrac{2i_0R - \xi}{4R} = 0$

Chemistry

PART – II

SECTION – A

19. B, C, D

20. B, C, D

21. A, B, C

22. A, B, C, D

23. B, C, D

Sol. The SRP should be high for a species to be a good oxidizing agent.

24. B, D

Sol. Factual

25. C

Sol. Boric acid (mild antiseptic), chloramphenicol (antibiotic) and aspirin (antipyretic as well as Analgesic) are all medicinal compounds.

26. B

Sol. N_2 : KK $\sigma_{2s}^2\sigma_{2s}^{*2}\ \pi_{2p_x}^2\ \pi_{2p_y}^2\sigma_{2p_z}^2$ (B.O = 3, Diamagnetic)

N_2^+ : KK $\sigma_{2s}^2\sigma_{2s}^{*2}\pi_{2p_x}^2\pi_{2p_y}^2\sigma_{2pz}^1$ (B.O = 2.5, Paramagnetic)

C_2 : KK$\sigma_{2s}^2\sigma_{2s}^{*2}\ \pi_{2p_x}^2\pi_{2p_y}^1$ (B.O = 2, Diamagnetic)

C_2^+ : KK $\sigma_{2s}^2\sigma_{2s}^{*2}\pi_{2p_x}^2\pi_{2p_y}^1$ (B.O = 1.5, Paramagnetic)

NO : KK $\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2p_z}^2\pi_{2p_x}^2\pi_{2p_y}^2\pi_{2p_z}^{*1}$ (B.O. = 2.5, Paramagnetic)

NO^+ : KK $\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2p_z}^2\pi_{2p_x}^2\pi_{2p_y}^2$ (B.O. = 3, Diamagnetic)

O_2 : KK $\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2p_z}^2\pi_{2p_x}^2\pi_{2p_y}^2\pi_{2p_x}^{*2}\pi_{2p_y}^{*1}$ (B.O. = 2., Paramagnetic)

O_2^+ : KK $\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2p_z}^2\pi_{2p_x}^2\pi_{2p_y}^2\pi_{2p_x}^{*1}$ (B.O. = 2.5, Paramagnetic)

27. B

28. B

SECTION – D

29. 00005.00

Sol. Given, $D_2 = 30.4\times10^{-7}$ cm

$D_1 = 108.5\times10^{-7}$ cm

Let excited state of He^+ be n_2. It comes from n_2 to n_1 and then n_1 to 1 to emit two successive photon

$$\frac{1}{D_2} = R_H.Z^2\left[\frac{1}{1^2} - \frac{1}{n_1^2}\right]$$

$$\frac{1}{30.4\times10^{-7}} = 109678\times4\left[\frac{1}{1^2} - \frac{1}{n_1^2}\right]$$

$\therefore n_1 = 2$

Now for $D_1 : n_1 = 2$ and n_2 : ?

$$\frac{1}{D_1} = R_H Z^2\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$$

$$\frac{1}{108.5\times10^{-7}} = 109678\times4\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$$

$\therefore n_2 = 5$

Thus, excited state for He is 5th orbit.

30. 00004.00

31. 00008.00

Sol. Total number of geometrical isomers in $\left[Pt\left(H_2N-CH(CH_3)-CO\overset{\ominus}{O}\right)_2\right]$ Total number of stereoisomer in $\left[Pt(gly)_3\right]^{\oplus}$ some of geometrical isomers and stereoisomers of given complexes = 8.

32. 00008.00

Sol. $\Delta T_f = \dfrac{1000\,K_f\,W_B}{M_B\,W_A}, M_B = 79x$

$$79x = \frac{1000\times5.12\times2.47}{0.4\times50} = 632.32 \Rightarrow x = \frac{632.32}{79} = 8$$

33. 00004.02

Sol.

$$\underset{2\,F}{Mg^{2+} + 2e^-} \longrightarrow \underset{24\,g}{Mg}$$

24 g Mg need change $= 2F = 2\times96500$ C

18 g Mg needs charge $= (2\times96500)\times\dfrac{18}{24}$

$It = Q \Rightarrow t = \dfrac{2\times96500}{10}\times\dfrac{18}{24}$ seconds

$= \dfrac{3474000}{240}$ seconds $= 14475$ seconds

$= \dfrac{14475}{3600}$ hr $= 4.02$ hours

34. 00002.82

35. 00086.20

Sol. Total mass of the water $= M = 100$ g Latent heat of vaporization of water at $0\ ^{\circ}C = L_1 = 21.0 \times 10^5$ J / Kg

Latent heat of fusion of ice $= L_2 = 3.36 \times 10^5$ J / Kg

Suppose, the mass of the ice formed $= m$

Then, the mass of water evaporated $= M - m$

Heat lost by the water in freezing = Heat taken by the water in evaporation

Thus, $mL_2 = (M - m)L_1$ or $m = 86.2$ g

36. 00008.00

Sol. In $Mg_xAl_yO_z$, O^{2-} ions are in ccp arrangement. Number of atoms in a unit cell of ccp $= 4$. Hence, there are 4 octahedral voids and 8 tetrahedra voids.

Total positive charge = charge on Mg^{2+} + charge on $Al^{3+} = \frac{1}{8} \times 8 \times 2 + \frac{1}{2} \times 4 \times 3 = 8$

Mathematics

PART – III

SECTION – A

37. B, C

Sol. Let $P(x) = 1 + x + \frac{x^2}{2!} + ... + \frac{x^{2018}}{2018!}$

$P'(x) = 1 + \frac{x}{1!} + ... + \frac{x^{2017}}{2017!}$

$\therefore \int (2018)! \frac{(P(x) - P'(x))}{P(x)} dx$

$= 2018! \int \left(1 - \frac{P'(x)}{P(x)}\right) dx$

$= 2018! (x - \ln P(x)) + C$

38. A, B, D

Sol. $a_n = \prod_{K=0}^{n} \left(1 + \frac{2}{x^{2^K} + x^{-2^K}}\right), |x| > 1$

$= \frac{x+1}{x-1} \cdot \frac{x^{2^{n+1}} - 1}{x^{2^{n+1}} + 1}$

$\lim_{n \to \infty} a_n = \left(\frac{x+1}{x-1}\right)\left(\frac{1-0}{1+0}\right) = \frac{x+1}{x-1}$

$\therefore f(x) = \frac{x+1}{x-1}$

39. A, B, C

Sol. $f(x)\left[f(x))^6 + 1\right] = x$

$f(0)\left[f(0)^6 + 1\right] = 0 \Rightarrow f(0) = 0$

& $7(f(x))^6 f'(x) = 1 - f'(x)$

$f'(x)\left[7(f(x))^6 + 1\right] = 1$

$\Rightarrow f'(x) > 0 \forall x \in R$

$\Rightarrow f(x)$ is increasing function $\forall x \in R$

$\Rightarrow x^7 + x = f^{-1}(x)$

$\Rightarrow \int_0^{\sqrt{2}} (x^7 + x)\, dx = 3$

& $\int_0^a f(x)dx + \int_0^{f(a)} f^{-1}(x)dx = af(a)$

$\Rightarrow \int_0^{\sqrt{3}} f(x)dx = \frac{f(\sqrt{3})}{8}\left[8\sqrt{3} - (f(\sqrt{3}))^7 - 4f(\sqrt{3})\right]$

40. A, C

Sol. Let T be the midpoint of PR, perpendicular from T to PR meets C_1 C_2.

Also $MP^2 = MR^2 = MQ^2 = MS^2 = MT^2 + TR^2$

Hence M is centre of required circle

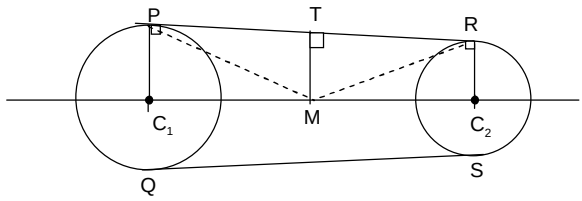
$PR = \sqrt{(C_1\ C_2)^2 - (r_1 - r_2)^2}$

$= \sqrt{100 - 4} \quad = \sqrt{96}$

$\lambda^2 = MT^2 + TR^2$

$= \left(\frac{2+4}{2}\right)^2 + TR^2 = 9 + \frac{96}{4}$

$= 9 + 24 = 33$



41. A, B, C, D

Sol. $S(n+1) = \sum_{r=0}^{n} (-1)^{n-r} \left\{1 - \frac{1}{r+1}\right\} \frac{(n+1)!}{r!(n+1-r)!}$

$= \sum_{r=0}^{n} (-1)^{n-r}\ {}^{n+1}C_r - \sum_{r=0}^{n} (-1)^{n-r} \frac{(n+1)!\ n+2}{(r+1)!\ (n+1-r)!} \cdot \frac{1}{n+2}$

$= 1 - \frac{1}{n+2} \sum_{r=0}^{n} (-1)^{n-r}\ {}^{n+2}C_{n+1-r}$

$= 1 - \frac{1+(-1)^n}{n+2}$

42. A, B, C, D

Sol. $|A(t)| = \frac{\sin 4t}{\sin t}$

$|A(4t)| = \frac{\sin 16t}{\sin 4t}.$

43. A

44. A

Sol. $(f(x))^2 = 1 + x\ f(x+1).....(1)$

$\frac{x+1}{2} \le f(x) \le 2\ (x+1).....(2)$

Replacing x by $x+1$in eqn (2)

$\frac{x+2}{2} \le f(x+1) \le 2(x+2).....(3)$

From (1) we have

$$\frac{1}{2} + x\ f(x+1) < (f(x))^2 < 2 + x\ f(x+1)$$

From (2) & (3)

$$\frac{1+x(x+2)}{2} < (f(x))^2 < 2\ [1+x(x+2)]$$

Taking square root

$$2^{-1/2}(1+x) < f(x) < 2^{1/2}\ (1+x)$$

On continuing the process, k times we get

$2^{-1/2^k}(1+x) < f(x) < 2^{1/2^k}(1+x)$

Let $k \to \infty$ gives $1+x \le f(x) \le 1+x)$

$\to\ f(x) = x+1$

45. A

46. C

Sol. $C_n = a_1 + a_2 + ... + a_n$

$a_1, a_2, ...a_n$ are in A.P with common difference = 2

$d_n = b_1 + b_2 + ... + b_n$ are in A.P with common difference = 2

(a_n, c_n) lies on $y = px^2 + qx + r$

$c_n = p\, a_n^2 + q\, a_n + r$

$c_{n-1} = p(a_{n-1}^2 + qa_{n-1} + r$

$c_n - c_{n-1} = p(a_n^2 - a_{n-1}^2) + q(a_n - a_{n-1})$

$\Rightarrow a_n = p(a_n + a_{n-1})(a_n - a_{n-1}) + q(a_n - a_{n-1})$

$a_n = (a_n - a_{n-1})\,[p(a_n + a_{n-1}) + q]$

let $n = 2$ & 3

$a_2 = d\,[p(a_2 + a_4) + q]$.....(1)

$a_3 = d\,[p(a_3 + a_2) + q]$.....(2)

Subtracting above 2 eqns (1) & (2)

$a_3 - a_2 = d\,[p(a_3 - a_1)]$

$4p = 1 \Rightarrow p = \frac{1}{4}$.

Adding (1) & (2)

$q = \frac{1}{2}$

SECTION – D

47. 00002.00

Sol. $f'(x) = -2 - 3x^2 < 0 \Rightarrow f(x)$ is decreasing

$\therefore\ f\big(f(x)\big) < f(-x) \Rightarrow f(x) > -x$

$\Rightarrow 30 - x - x^3 > 0 \Rightarrow x^3 + x - 30 < 0$

$\Rightarrow\ (x-3)(x^2 + 3x + 10) < 0$

$\Rightarrow\ x < 3$

No. of values = 2

48. 00010.00

Sol. Area of Δ ABC $\Rightarrow \frac{1}{2}ab = 1$

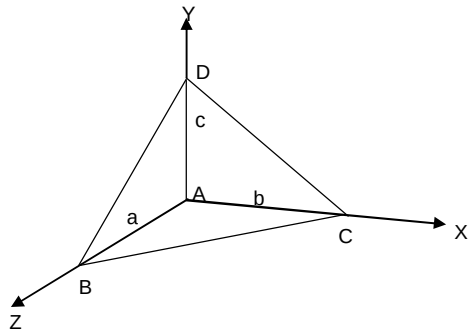
Area of Δ ACD $= \frac{1}{2}bc = 3$

Area of Δ ADB $= \frac{1}{2}ac = \sqrt{90}$

Area of ΔBCD

$= \frac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2}$

$= \frac{1}{2} \times 2\sqrt{1 + 9 + 90} = 10$



49. 00005.00

Sol. On subtracting given equation

$\left[(\tan p - 1)^3 - (1 - \cot q)^3\right] + 2019\left[(\tan p - 1) - (1 - \cot q)\right] = 0$

$\Rightarrow\ [\tan p + \cot q - 2]\,[(\tan p - 1)^2 + (1 - \cot q)^2 + (\tan p - 1)\,(1 - \cot q) + 2019] = 0$

This is possible if $\tan p + \cot q = 2$

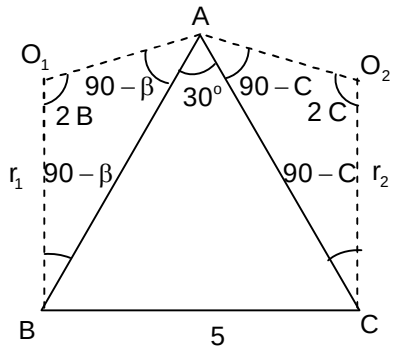
$\therefore \sin r + \cos r = 1$

$\Rightarrow r = -2\pi, -\frac{3\pi}{2}, 0, \frac{\pi}{2}, 2\pi$

50. 00025.00

Sol. In $\Delta AO_1B \; \frac{AB}{\sin 2B} = \frac{r_1}{\cos B}$.....(1)

In $\Delta AO_2B \; \frac{AC}{\sin 2C} = \frac{r_2}{\cos C}$.....(2)



From above (2) equations

$\frac{AB \,.\, AC}{\sin 2B \,.\, \sin 2C} = \frac{r_1 \, r_2}{\cos B \cos C}$

$\frac{2R \sin C \,.\, 2R \sin B}{2 \sin B \; 2 \sin C} = r_1 \, r_2$

$r_1 \, r_2 = R^2 = \left(\frac{5}{2 \sin 30^o}\right)^2 = 25$.

51. 00153.86

Sol. $D \le 0$

$4(p + q - 7)^2 - 8pq \le 0$

$\Rightarrow (p-7)^2 + (q-7)^2 \le 7^2$ interior & circumference of this represents an circle with centre (7, 7) & radius 7

$\therefore$ Area $= 49\pi$.

52. 00018.00

Sol. $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 2 \end{vmatrix} [\bar{p} \; \bar{q} \; \bar{r}]$

$= -18(\bar{p} \times \bar{q}) \,.\, \bar{r} = -18 \, \bar{r}^{\,2}$

So maximum value of $\left|-18\bar{r}^{\,2}\right|$ is 18.

$\left(\because |\bar{r}| \le |\bar{p}| \, |\bar{q}|\right)$

53. 00009.00

Sol. Required Probability $= \frac{{}^3C_1 . {}^3C_2}{{}^7C_4} = \frac{9}{35}$

54. 00007.00

Sol. $\sum_{n=1}^{1000} n^{-2/3} > \int_1^{1000} x^{-2/3} dx = 3(x^{1/3})^{1000} = 27$.....(1)

& $\sum_{n=1}^{1000} n^{-2/3} < \int_1^{1000} x^{-2/3} \, dx + \left\{\left(1^{-2/3} - 2^{-2/3}\right) + \left(2^{-2/3} - 3^{-2/3}\right) + \right.$

$\left.(3^{-2/3} - 4^{-2/3}) + ... + (999^{-2/3} - 1000^{-2/3})\right\} + (1000)^{-2/3} = 28$ ..... (2)

$\therefore$ From (1) & (2)

$M = \left[\sum_{n=1}^{1000} n^{-2/3}\right] = 27$

$\Rightarrow M - 20 = 7$